REFLOWECONOMY · CALCULATION NOTE

ReflowEconomy reference micro-factory sizing and economics

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade and STEM check (RFE-STM-001): motor energy at the electrical input (45.6 kWh, 0.88 kWh/kg; RFE-DEC-001 D2); supply counted as currents (block A about 55 A against 40 A; D1 to Amish); feed slot 120 mm so ISO 13857 Table 4 applies (D3); result +$0.17 per shift; results reported against the PR numbers of RFE-PRS-001
0.5	2026-10-02	Amish Chadha	Approved follow-ups: ISO 13857 reach check done (straight collar fails, cranked feed hood passes at 1,060 mm); headroom 2.02 m under the padded tray crossing; escape widths past the open door and gates; equipment $25,505; result +$0.57 per shift
0.4	2026-10-02	Amish Chadha	Guarding note: ISO 13857 reach check from the chute mouth done now (decided by Amish, 2026-10-02)
0.3	2026-09-30	Amish Chadha	

			Design for construction (RFE-DDR-003): equipment $25,300 against the R10 value-engineering target of $25,000; result +$0.73 per shift; aisle 1.57 m; seam beam space and aisle headroom
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Fume fan 1.5 kW (maximum demand 8.75 kW, 38.0 A; equipment $24,200; result +$1.61 per shift); intake quality rule (R16, 10 % residue threshold) added to the mass balance; rented shed preferred over containers
0.1	2026-09-25	Amish Chadha	First issue at TRL 3 (mass balance, throughput, energy, supply, fume hoods, noise, layout, cost, economics model, passport v0.2 check)

ReflowEconomy reference micro-factory sizing and economics

On paper, the reference micro-factory handles 100 kg of mixed input per 8 h shift on 59.4 m², keeps 51.6 % of the input local as products and clean flake, uses 0.88 kWh per kilogram of output with motor and drive losses counted, and costs $25,505 in equipment ($505 over the PR-010 (R10) value-engineering target of $25,000). Counted as currents, its morning block of loads draws about 55 A from the 230 V, 40 A single-phase supply, so the supply requirement PR-017 is **not met** and the way to meet it is proposed to Amish (RFE-DEC-001 D1). It breaks even at best: the result is about +$0.17 per shift before rent at the reference prices, and the break-even product price is $2.49/kg against an assumed $2.50/kg. Four requirements are **not met** (PR-001, PR-004, PR-008 and PR-017; for PR-008, 28.4 % of the reference input goes to licensed disposal against a 20 % target, and the intake quality rule PR-016 shows a load needs 10.6 % residue or less to meet it), two are **at risk** (PR-006 throughput, PR-012 noise) and three cannot be verified at TRL 3 (PR-011 exposure, PR-015 form time, PR-020 registration). Requirement numbers are those of RFE-PRS-001; PR-001 to PR-016 were R1 to R16.

Every number in this note is printed by `python docs/04-calcs/sizing.py`, which also writes `docs/04-calcs/results.csv`. The script reads the layout from `cad/src/model.py`, the equipment cost from `bom/bom.csv` and the economics from `docs/playbook/economics_model.py` with `docs/playbook/economics_inputs.csv`, and it stops with an error if the economics inputs disagree with this calculation. All values are estimates for a paper proof of concept; none has been measured.

> **Safety:** This note sizes guarding, fume extraction and electrical supply on paper only. The micro-factory combines a shredder, surfaces at 190 to 200 °C, melt fumes, water near electricity and a high fire load. A qualified safety professional and a licensed electrician must review the design before any site operates.

1. Basis

Decisions applied (RFE-DDR-001, decided by Amish on 2026-09-25): plastics-only reference line, with aluminium as a separate add-on bay (item 1); two 40 ft container footprints (item 2); PET sold as clean flake (item 3); passport schema v0.2 (item 4); single-phase supply with staggered heating (item 5). Further decisions accepted by Amish on 2026-09-25 (RFE-DDR-002): R8 keeps counting process losses and an intake quality rule is added (item 8); the derated 5 kW single-phase press stays the reference, with the published 15 kW press where 400 V exists (item 9); a 1.5 kW fume fan (item 10); a rented shed is preferred to the container option (item 11).

Shift pattern (assumption): one 8 h shift, four workers. Block A (hours 0 to 4) is sorting, washing, shredding and extrusion; block B (hours 4 to 8) is the sheet press. The hot zone runs on flake dried during the previous shift, so extrusion can start at once.

2. Mass balance

Assumptions: source-separated or picker-sorted post-consumer input of 22 % PET, 20 % HDPE, 18 % PP (60 % target plastic), 8 % metals, 2 % e-waste, 10 % paper and card and 20 % residue by mass; washing and float-sink remove 10 % of plastic mass; shredding loses 1.5 % and drying 1 % as fines and spills; 3 % of the melt feed is lost as purge and degraded material; trim of 8 % of the gross melt output is re-shredded and returned. Dry basis.

Table 1. Mass balance per 100 kg of input (estimates).

Stream	Mass	Destination
HDPE and PP products (sheets, beams)	32.3 kg	Local sale with passport
PET clean flake	19.3 kg	Local or regional sale with passport
Metals (aluminium, steel)	8.0 kg	Sold for smelting, with passport
E-waste boards and cells	2.0 kg	Licensed refiner or cell tester, with passport
Paper and card	10.0 kg	Local mill
Sorting residue (PVC, multilayer, film, organics)	20.0 kg	Licensed disposal, never burned
Process losses (labels, dirt, fines, purge)	8.4 kg	Licensed disposal

The mass closes at 100.00 kg. Products and flake kept local are **51.6 %** (R7, target 50 %: met with a margin of 1.6 points). Everything sent to licensed disposal is **28.4 %** (R8, target 20 %: **not met**). The sorting residue alone is exactly 20 %; the process losses push the total over. R8 depends mostly on collection quality, which the site does not control directly.

Intake quality rule (R16). Amish decided on 2026-09-25 (RFE-DDR-002, item 8) to keep the honest total in R8, process losses included, and to add an intake quality rule. For a load with sampled residue r, the other streams scale with (100 - r), so the process losses are 8.4 x (100 - r) / 80 kg per 100 kg. Disposal stays at 20 % or less when r is **10.6 %** or less. The rule threshold is set at **10 %** residue by sampled mass: at that threshold disposal is 19.5 % and 58.0 % of input is kept local. Loads above it are refused or charged a disposal fee. A load at the threshold carries more plastic, 67.5 kg instead of 60 kg, so the shredder load rises to 63.9 kg, which needs 4.26 h at 15 kg/h against the 4 h block: meeting PR-008 (R8) this way makes PR-006 (R6) worse (found by the independent review of the STEM check, 2026-10-25). The reference mix (20 % residue) would fail the rule, so R8 stays **not met** for the reference input; meeting it depends on a cleaner supply agreed with collectors, which only a co-design partner can confirm. The TRL 2 precis gave 30 kg of products, 20 kg of flake and 10 kg of losses; those figures are replaced by Table 1.

3. Throughput and time budget

Assumptions: two sorters at 25 kg/h each; shredder effective rate 15 kg/h, within the 8.8 to 41.8 kg/h range published for the Precious Plastic Shredder Pro (Precious Plastic Academy); extruder 5 kg/h of beams; sheet press cycle 55 min per 12 mm sheet, as published for the Precious Plastic sheetpress (Precious Plastic Academy); heaters off 30 min before the end of the shift.

Table 2. Time budget per shift.

Step	Load	Time needed	Time available	Margin
Sorting	100 kg	2.0 h	4 h (block A)	Ample
Shredding	56.8 kg (54.0 kg washed plastic plus 2.8 kg trim)	3.79 h at 15 kg/h	4 h	5 %; needs 14.2 kg/h
Extrusion	12.3 kg of beams	2.79 h including 0.33 h heat-up	4 h	30 %
Sheet press	2 sheets of 11.4 kg	1.62 h heat-up plus 2 x 55 min	3.5 h	2.06 sheets of capacity for 2
Drying	52.7 kg of flake	7.3 kg/m² on 7.2 m² of trays	10 kg/m²	27 %

R6 (100 kg or more per shift) is **at risk**. The shredder has 5 % margin at an assumed rate; at the low end of the published range (8.8 kg/h) it would need 6.5 h. The sheet press has almost no margin because heating 285 kg of steel platens from cold on 5 kW takes 1.62 h. A 5 kW single-phase press is a derated variant: the published Precious Plastic sheetpress is 15 kW on a 400 V, 32 A supply, and the Shredder Pro and Extrusion Pro are also specified for 400 V. Their motors can run on single-phase-input variable-frequency drives; the press heaters cannot deliver 15 kW on a 40 A single-phase supply. Amish decided on 2026-09-25 (RFE-DDR-002, item 9) that the derated 5 kW press stays the playbook reference, with the published 15 kW press as an option where a 400 V supply exists; the 15 kW press would heat up in well under an hour and remove the press constraint, but it needs three-phase. The rates must be measured on the chosen machines.

4. Energy per shift

The sheet press is sized from first principles. Heat-up: two 1.1 x 1.1 m steel platens 15 mm thick (285 kg, specific heat 0.49 kJ/kg K) from 25 to 200 °C take 6.79 kWh. Per sheet: 11.4 kg of HDPE (specific heat 2.25 kJ/kg K over 175 K plus 180 kJ/kg latent heat) and 31 kg of steel mould take 2.56 kWh. The standing loss of the hot press is taken as 0.8 kW (assumption). During cycles the press draws about 3.6 kW of its 5.0 kW, so the heaters can hold the published 55 min cycle.

Motor ratings are shaft output, so the electricity a motor draws is its shaft power divided by the motor and drive efficiencies (STEM check fix, RFE-DEC-001 D2, decided under Amish's standing instruction of 2026-10-23). The efficiencies are the IE3 minimum values of IEC 60034-30-1 for four-pole motors at 50 Hz, 86.7 % at 2.2 kW and 85.3 % at 1.5 kW (ABB technical note, 2018, checked 2026-10-25), with a drive efficiency of 96 % (assumption) and 75 % for the single-phase pump motor (assumption, below the IE2 value of 79.6 % at 0.75 kW). The drying fan's 0.4 kW is taken as its electrical rating.

Table 3. Energy per shift (estimates).

Load	Basis	Energy
Shredder	2.2 kW at 70 % for 3.79 h, over 0.867 x 0.96	7.0 kWh
Washing pump	0.75 kW for 4 h, over 0.75	4.0 kWh
Drying fan	0.4 kW for 8 h	3.2 kWh
Extruder	2.0 kW heat-up for 0.33 h; 2.0 kW heaters at 60 % duty and 1.5 kW motor at 60 % (over 0.853 x 0.96) for 2.45 h	6.3 kWh
Sheet press	Heat-up at 5 kW for 1.62 h; 2 sheets at 2.56 kWh; 0.8 kW loss over 2 cycles	14.7 kWh
Fume extraction	1.00 kW input for 7.2 h (1.5 kW fan)	7.2 kWh
Lighting, desk and small loads	0.4 kW for 8 h	3.2 kWh
Total		**45.6 kWh**

The line uses **0.88 kWh per kilogram** of products and flake (51.6 kg), against the PR-009 (R9) target of 1.0 kWh/kg: met on paper. On the old shaft basis it was 42.9 kWh and 0.83 kWh/kg. Not counted: hot wash water, a dewatering centrifuge, conveyors and a compressor, none of which is in the reference line; adding any of them reduces the margin to the 1.0 kWh/kg target. Of the total, 22.4 kWh per shift is fixed (heat-up, fume fan, lights and drying fan) and does not fall if less material comes in.

5. Electrical supply

Amish decided on single-phase supply with staggered heating (RFE-DDR-001, item 5). The supply is taken as 230 V, 40 A (9.2 kW). The staggering rule is enforced by an interlock contactor on the electrical board: the press heaters are locked out while the shredder or the extruder runs.

Table 4. Peak demand on the power basis (superseded by Table 4a).

Block	Loads running together	Peak
A (sort, wash, shred, extrude)	Shredder 2.2, pump 0.75, drying fan 0.4, lights 0.4, extruder 3.5, fume fan 1.5 kW	8.75 kW
B (press)	Press heaters 5.0, drying fan 0.4, lights 0.4, fume fan 1.5 kW	7.30 kW

Versions 0.1 to 0.5 of this note divided 8.75 kW by 230 V and gave 38.0 A, 5 % under the supply. That treats each motor's shaft rating as if it were drawn from the supply at unity power factor. The STEM check (RFE-STM-001 S14) counts currents instead: a drive with a single-phase diode input draws a peaky current about 2.4 times its rating divided by the voltage, and a single-phase motor on line draws more than its rating for its efficiency and power factor.

Table 4a. Input current in block A (estimates; running loads as in section 4).

Load	Basis	Full load	Running
Shredder drive, 2.2 kW	20.2 A at 240 V for a 2.2 kW single-phase input drive (Schneider ATV12HU22M2), scaled to 230 V; 70 % load	21.1 A	14.8 A
Extruder drive, 1.5 kW	14.9 A at 240 V (Schneider ATV12HU15M2); 60 % load	15.5 A	9.3 A
Extruder heaters, 2.0 kW	Resistive	8.7 A	8.7 A
Fume fan, 1.5 kW single-phase	2 hp, 12.0 A at 230 V (NEC Table 430.248, as reproduced by Elliott Electric); running at 1.00 of 1.5 kW	12.0 A	8.0 A
Washing pump, 0.75 kW	1 hp, 8.0 A (NEC Table 430.248)	8.0 A	8.0 A
Drying fan, 0.4 kW	1/2 hp, 4.9 A (NEC Table 430.248)	4.9 A	4.9 A
Lighting, 0.4 kW	Resistive	1.7 A	1.7 A
Block A		**72.0 A**	**55.4 A**

The block A load is about **55 A running and 72 A at full load against the 40 A supply**: PR-017 is **not met** (all sources checked 2026-10-25). The figures add table currents arithmetically, so they are a conservative upper estimate (the fan's table current, for example, is above what a modern fan draws); even well below them block A exceeds the supply. Block B runs at 91 % of the supply for about 3.5 h, and a common practice limits continuous loads to 80 % of the breaker rating, so the electrician may also find block B too close. The MSEDCL source is the 2010 Conditions of Supply; the current supply code is to confirm. Block B is 36.4 A running (40.4 A at full load), within the supply. Without the interlock the connected load would be far higher still. Motor starting currents are not included; the shredder and extruder start through their drives.

The supply cannot be fixed in this note without changing an input Amish decided, so the options are proposed, awaiting Amish (RFE-DEC-001 D1):

- **(a) A 400 V three-phase supply.** The same loads shared over three phases are about 18.5 A a phase if they are balanced. In Maharashtra, the first partner region, single-phase supply is given only up to 40 A and anything above is a three-phase connection (MSEDCL Conditions of Supply, clause 6.4.1, checked 2026-10-25). Three-phase also allows the published 15 kW sheet press (section 3). Recommended.
- **(b) Single phase with three blocks.** Sort, wash and shred (29.4 A), then extrude (32.6 A; the fume fan runs only while a heater is on), then press (36.4 A). The working day grows to about 10.6 h, which adds about 12.9 USD a shift in wages at the reference wage and turns the result per shift from +0.17 to about -12.7 USD.
- **(c) Single phase, 8 h, less input.** The throughput target PR-006 would drop below 100 kg a shift.

The licensed electrician must confirm the supply rating, earthing and RCD selection on site, with the input currents of the drives actually bought (RFE-DEC-001, item 9 to confirm).

6. Fume extraction

R11 asks for a face velocity of 0.5 m/s or more at the hood of every melt process. A canopy over the whole hot zone (2.0 x 3.5 m) would need 3.5 m³/s and about 6.1 kW of fan power, more than the single-phase supply can spare, so the design uses two enclosing hoods:

- **Hood A**, a booth over the sheet press and its cooling press, with a 1.2 x 0.6 m sliding sash for loading.
- **Hood B**, an enclosure over the extruder barrel with a 0.8 x 0.4 m open face at the nozzle and mould.

Table 5. Hood and fan sizing.

Quantity	Value	Basis
Open face area	1.04 m²	0.72 m² (A) plus 0.32 m² (B)
Air flow	0.572 m³/s (2,059 m³/h)	0.5 m/s plus 10 % for enclosure gaps
Main duct	250 mm, 11.7 m/s, velocity pressure 81 Pa	Galvanized steel
Duct friction	47 Pa	Friction factor 0.018 over 8 m equivalent
System pressure	815 Pa	Hood entry 1.5 VP, fittings 1.4 VP, stack 1 VP, filters dirty: 250 Pa (G4 plus F7) and 200 Pa (carbon)
Fan input power	1.00 kW (66 % of the 1.5 kW rating)	Fan 55 % times motor 85 %
Room air changes	15 per hour in 136 m³	Make-up opening of 0.38 m² at 1.5 m/s

Contaminants and limits. Processing polyolefins releases hydrocarbons, formaldehyde, acetaldehyde, formic acid and acetone, carbon monoxide and a wax-like aerosol (Borealis, *Statement on impurities: polyolefins*, ed. 6, 2022). The playbook uses the lower of the local limit and the following US OSHA permissible exposure limits as a floor: formaldehyde 0.75 ppm over 8 h and 2 ppm over 15 min (NIH summary of 29 CFR 1910.1048); acrolein 0.1 ppm; acetaldehyde 200 ppm (OSHA Table Z-1). More protective limits exist (for example the ACGIH ceiling for formaldehyde), and a site should adopt them where it can.

Filters. A G4 prefilter and an F7 bag filter catch the wax aerosol; an impregnated activated carbon stage takes organic vapors. Plain activated carbon holds formaldehyde poorly, which is one reason the fan and filter sit outside and discharge through a stack above the roof, away from doors and air intakes.

R11 status: the hoods meet 0.5 m/s by design, and PVC, polystyrene and unknown plastics are excluded by the recipes. Whether air at the workstations stays below the exposure limits can only be shown by air monitoring on a running site, so R11 is **not verifiable at TRL 3**. Amish decided on 2026-09-25 (RFE-DDR-002, item 10) to fit a 1.5 kW fan, because filter loading is hard to police on a small site: with dirty filters it runs at 66 % of its rating instead of 91 % with the 1.1 kW fan. The block A peak on the power basis rose from 8.35 kW to 8.75 kW; section 5 now counts currents instead.

7. Shredder noise and guarding

Assumptions: the bare shredder chopping rigid plastic has an A-weighted sound power level of 102 dB (a typical value for small granulators, not a measurement); the room is the steel shell (surface 194 m², mean absorption coefficient 0.1, room constant 21.5 m²); the source sits on the floor (directivity 2); the operator is 1 m away and other workers 4 m away; the shredder runs 3.79 h per shift.

Table 6. Noise at the workstations.

Case	1 m	4 m	Operator LEX,8h
Bare shredder	97.4 dB(A)	94.9 dB(A)	94.1 dB(A)
In a lined enclosure (15 dB insertion loss)	82.4 dB(A)	79.9 dB(A)	79.1 dB(A)

The steel room is reverberant, so a bare shredder puts the whole floor above 85 dB(A), not only the shredder zone. With the lined acoustic enclosure the operator exposure is 79.1 dB(A) over 8 h, below the 85 dB(A) limit in R12. With a 15 dB enclosure, the design still meets 85 dB(A) for any bare-machine sound power up to 107.9 dB; both the sound power and the insertion loss are assumptions. The shredder area stays a hearing protection zone until the level is measured.

Guarding (defined, not detailed): fixed guards on all drive parts; feed through an interlocked chute in the enclosure roof, so the rotor cannot be reached with the lid open; interlocked enclosure door; emergency stop at the chute and at the board; lockout before clearing jams. Opening sizes and reach distances are checked against ISO 13857. Decided by Amish on 2026-10-02: the reach from the 1.70 m chute mouth to the cutters is checked from the model now, not at the detailed design stage or the safety review, and the chute is lengthened or baffled until it passes.

Reach check (ISO 13857, from the distances in `cad/src/model.py`). The top of the cutting chamber is taken as 1,100 mm above the floor, about 410 mm below the top of the 1,512 mm machine envelope; this is an assumption to be measured on the chosen machine. ISO 13857 Table 4 gives a safety distance of 850 mm for slot openings from 40 to 120 mm (the arm can reach in up to the shoulder), measured along the shortest path a hand can take; for openings over 120 mm the standard sends the designer to the reaching-over distances instead (Troax safety guide; Axelent, checked 2026-10-25), and slots over 180 mm admit the whole body. Versions 0.4 and 0.5 of this note applied the 850 mm distance to a 160 mm slot, which the table does not cover; the STEM check (RFE-STM-001 S17) found it, and the slot is now 120 mm (RFE-DEC-001 D3, decided under Amish's standing instruction of 2026-10-23).

Table 6a. Reach from the feed mouth to the cutters.

Case	Path from the mouth to the cutters	ISO 13857 distance	Result
Straight collar, 460 x 370 mm mouth facing up at 1.70 m	600 mm straight down	850 mm	**Fails** by 250 mm, and the opening admits the head
Cranked feed hood on the collar: 460 x 160 mm slot facing the aisle, lip at 1.70 m (v0.5)	460 mm along the hood floor, then 600 mm down: 1,060 mm	Not covered by Table 4 (slot over 120 mm)	**Fails** the standard's own scope
Cranked feed hood with a 460 x 120 mm slot (v0.6), lip at 1.70 m	460 mm along the hood floor, then 600 mm down: 1,060 mm	850 mm (Table 4, 40 < e <= 120 mm)	**Passes** by 210 mm

The straight collar fails, so the chute is baffled as decided: a cranked feed hood with a closed top sits on the collar, and the feed enters through a 120 mm slot in its aisle-side end with the bottom lip still at 1.70 m. The slot carries the interlocked flap that replaces the lid. Feed pieces must be thinner than 120 mm: bottles are crushed flat and bigger parts cut at the sorting table. The result rests on the assumed cutter height: each 100 mm higher takes 100 mm off the 210 mm margin, so the check is repeated with the chosen machine's drawing, and the safety professional confirms it (stop S7).

PR-012 (R12) status: **at risk**. The noise result rests on an assumed sound power and insertion loss, and the guarding is defined but not detailed; the feed reach check passes on paper. The insulated hot surfaces and 30 mA RCDs are specified in the BOM.

8. Layout checks

The layout is built by `cad/src/model.py` and drawn on RFE-DWG-001 Rev P4.

Table 7. Layout checks against R5.

Check	Value	Target
Footprint	12.19 x 4.88 m = 59.4 m² (56.7 m² inside)	60 m² or less
Central aisle	1.57 m clear between equipment; 1.20 m painted; 2.04 m headroom under the cable tray crossing, 2.02 m under its padding	1.0 m or more; headroom 2.0 m or more
Escape width with a door open	1.66 m past the open enclosure door; 1.08 m past the open cage gates	Confirmed by the safety professional against the local code (stop S7)
Hot zone clearance to stock	1.00 m (to racking bay 1)	1.0 m or more
Exits	Intake and export doors at X = 0, personnel exit at the far end; worst travel along the aisle 7.9 m	Two exits
Zone overlaps	None	None

R5 is **met**, with the hot zone clearance exactly at its limit. Amish decided on 2026-09-25 (RFE-DDR-002, item 11) that a rented shed of the same size is the preferred shell where one is available. The container option is kept for sites without a shed; removing the adjoining side walls of two containers weakens them, so it is done only to a structural engineer's design. The constructable design (RFE-DDR-003) reserves a 165 x 305 mm space under the roof along the seam for the engineer's seam beam; the lowest thing over the aisle is the cable tray crossing, 2.04 m above the floor. As decided by Amish on 2026-10-02, the crossing is padded with 15 mm closed-cell foam over the aisle and marked with yellow and black hazard tape; the underside of the padding is 2.02 m above the floor, still above the 2.0 m check. The enclosure door (790 mm leaf) and the cage gates (590 mm leaves) open into the aisle only during locked-off maintenance or loading and close themselves on spring hinges; with them open, 1.66 m and 1.08 m of aisle remain, so hinged doors are kept (the decision calls for sliding doors only if the escape width falls short).

Detailed model (kit 1.9). The layout above is the envelope level that this note and the drawings read. The same layout is also modelled at detail level D3 in `cad/src/detail_a.py` (`components("A")` in `cad/src/model.py`): every made part from standard stock (angle, SHS, RHS, flat bar, sheet, mesh, plywood) with its bolts, screws, anchors and weld beads, which is the evidence for PR-018 (local fabrication), and the cell container, extinguishers

and the 1.0 m stock clearance for PR-019 (fire and cells). The fidelity check `python .kit/cad_check.py --export` passes at D3 with no interference between parts (`cad/fidelity.json`). The container floor carries the 430 kg hot press at 0.36 t/m against 3.0 t/m allowed (RFE-STM-001 S19), and a UB 305 x 165 x 40 fits the space reserved for the seam beam and carries the roof edges with its top flange held by the roof (S18); the structural engineer designs the real beam.

9. Equipment cost

`bom/bom.csv` prices every line. Items 1 to 12 total **$25,505** against the R10 value-engineering target of $25,000 (a hypothetical control target, not a limit): $505 over the target. The building shell adds about $9,200 for the container option. Against the TRL 2 list, the shredder line gained a $600 acoustic enclosure, the fume line $300 for a second hood and the carbon stage, and the board $300 for the heater interlock. In v0.2 the fume line gains a further $150 for the 1.5 kW fan (RFE-DDR-002), which moves the total from $24,050 to $24,200. In v0.3 the design for construction (RFE-DDR-003, 2026-09-30) adds $1,100 to items 1 to 12 (drying fan stand $50, transfer bridge $250, sash, take-off, brackets, wall sleeve and fan stand $400, backboard and cable tray $400) and $1,200 to the container shell (seam beam, floor seam plate, roof flashing). In v0.5 the approved follow-ups of 2026-10-02 add $205: the cranked feed hood and three self-closing door hinges ($105, item 3), the tray crossing padding and hazard tape ($40, item 11) and two pairs of self-closing gate hinges ($60, item 12). `budget_usd` in `project.yaml` stays null because this is a playbook repo; R10 is the value-engineering target instead.

10. Economics model

The operator-editable model is `docs/playbook/economics_model.py` with its inputs in `docs/playbook/economics_inputs.csv` . An operator copies the CSV, changes the values to local prices, wages, tariff and yields, and runs `python docs/playbook/economics_model.py my_inputs.csv` . The model scales product, flake, metal, paper and disposal masses with input, splits energy into a fixed part (22.4 kWh per shift) and a part that scales with input, and treats wages, consumables, filters, rent and equipment recovery as fixed per shift. The reference inputs are illustrative and must come from the co-design partner.

Table 8. Result per shift at the reference inputs (USD, estimates).

Item	Basis	Amount
Products	32.3 kg at $2.50/kg	+80.75
PET flake	19.3 kg at $0.60/kg	+11.58
Metals	8.0 kg at $0.50/kg	+4.00
Paper and card	10.0 kg at $0.05/kg	+0.50
E-waste	2.0 kg at $0 until a receiver quotes	0.00
Feedstock	100 kg at $0.15/kg	-15.00
Wages	4 workers at $10	-40.00
Energy	45.6 kWh at $0.15/kWh	-6.84
Disposal	28.4 kg at $0.05/kg	-1.42
Consumables	Allowance	-10.00
Filters	Allowance	-3.00
Operating margin		**+20.57**
Equipment recovery	$25,505 over 5 years of 250 shifts	-20.40
Result before rent and finance		**+0.17**

Break-even product price: **$2.49/kg** at 100 kg input (the reference is $2.50/kg). Break-even input: **100 kg per shift**, the line capacity. Each $0.50/kg on the product price moves the result by $16.15 per shift, far more than any machine choice. The full cost of products and flake is $1.87/kg of output. PR-003 (R3) is **met**: the model exists, works per kilogram and per shift, gives the break-even points and is editable. Its review with an operating recycler is later work.

11. Material passport

Schema v0.2 (`standards/material-passport-v0.2.schema.json`) adds the fields Amish approved: `schema_version` , `product_form` , `parent_batch_ids` and an ISO 14021 aligned `recycled_content` statement. Schema v0.1 (`standards/material-passport.schema.json`) is kept unchanged. The v0.2 schema requires `parent_batch_ids` for every lot that is not an intake batch and a recycled content statement for flake, sheet, beam and profile lots. Following ISO 14021, rework or regrind reclaimed within the same process (the sheet trim here) is not counted as pre-consumer material (ISO 14021 definitions as quoted in a published declaration).

The script validates the three example records in `standards/examples/` (an intake batch, an HDPE sheet lot and a circuit-board export lot): 3 of 3 are valid against v0.2, and 3 of 3 are also valid against v0.1. A sheet lot without `parent_batch_ids` is rejected. R13 and R14 are **met** on paper. The example values are invented.

Validation limits. The passport check uses three invented example records and one negative case; it shows the schema behaves, not that every real lot will be valid.

Registration (PR-020). At 100 kg a shift and 250 shifts a year the reference site processes about 25 t a year, which puts it in the lowest fee band (under 200 t a year) for registration as a plastic waste processor with the Maharashtra Pollution Control Board on the CPCB portal, alongside consents under the Air and Water Acts (RFE-REG-001). Registration needs a real site and its documents, so PR-020 is verified at TRL 4 or later.

12. Results against requirements

Table 9. Product requirement status at TRL 3 (from `docs/04-calcs/results.csv`; PR numbers of RFE-PRS-001, old R numbers in brackets).

ID	Target	Value	Status
PR-001 (R1)	Feasibility matrix: 7 groups with route, rating, hazard, export point and a source per row	7 groups with route and reason; no hazard, export point or source columns	**Not met**
PR-002 (R2)	Sourced recipe for PET, HDPE, PP and aluminium	4 recipes in `docs/playbook/recipes/` (aluminium as the add-on bay)	Met
PR-003 (R3)	Operator-editable economics model with break-even	Model and CSV; result +$0.17 per shift; break-even product price $2.49/kg	Met
PR-004 (R4)	Every export stream names refining step, receiver type and packing rule	Principle and cell packing rule only	**Not met**
PR-005 (R5)	60 m² or less, aisles 1.0 m or more, separate hot zone	59.4 m²; aisle 1.57 m; headroom 2.02 m under the padded tray crossing; hot zone 1.00 m from stock	Met
PR-006 (R6)	100 kg or more of mixed input per 8 h shift	Shredder 3.79 h of 4 h at an assumed 15 kg/h; press capacity 2.06 sheets for 2	At risk
PR-007 (R7)	50 % or more kept local as products or clean flake	51.6 %	Met
PR-008 (R8)	Residue to licensed disposal 20 % or less, process losses counted	28.4 % at the reference mix (20 % sorting residue plus 8.4 % process losses); 19.5 % for a load at the PR-016 threshold	**Not met**
PR-009 (R9)	1.0 kWh/kg of output or less, motor and drive losses counted	0.88 kWh/kg	Met
PR-010 (R10)	Equipment items 1 to 12: value-engineering target $25,000	$25,505 ($505 over the target)	Over target
PR-011 (R11)	0.5 m/s on every melt process; no PVC, PS or unknown plastics; air below exposure limits	Hoods sized for 0.5 m/s (0.57 m³/s, fan 1.00 kW); exposure needs air monitoring	Not verifiable at TRL 3
PR-012 (R12)	Guarded shredder, insulated hot surfaces, RCDs, 85 dB(A) LEX or hearing zones	Guarding defined; feed reach 1,060 mm through a 120 mm slot against 850 mm (ISO 13857 Table 4, passes); LEX 94 dB(A) bare, 79 dB(A) enclosed (sound power assumed)	At risk
PR-013 (R13)	Valid passport for every outgoing lot	3 of 3 examples valid against schema v0.2	Met
PR-014 (R14)	Passport lists parent batch IDs and schema version	v0.2 requires both for every non-intake lot	Met
PR-015 (R15)	Passport filled in 2 min or less	Needs a form and a timed trial	Not verifiable at TRL 3
PR-016 (R16)	Intake quality rule with a residue threshold that keeps PR-008 within 20 %	Threshold 10 % residue by sampled mass (limit 10.6 %); loads above it refused or charged	Met
PR-017		Block A 55 A running (72 A at full load); block B 36 A	**Not met** (RFE-DEC-001 D1)

ID	Target	Value	Status
	Loads that run together within a 230 V, 40 A single-phase supply, counted as currents		
PR-018	Made parts from standard stock with workshop tools	Detailed model at D3 from standard stock; 11 made components with making sketches (section 8)	Met by design
PR-019	Cells out at intake into dry sand; three extinguishers; stock 1.0 m from the hot zone	All in the model and BOM lines 10 and 12 (section 8)	Met by design
PR-020	Registrations and consents listed with the site's capacity band	About 25 t a year, under 200 t a year band (section 11; RFE-REG-001)	Not verifiable at TRL 3

Summary: 4 not met (PR-001, PR-004, PR-008, PR-017), 2 at risk (PR-006, PR-012), 3 not verifiable at TRL 3 (PR-011, PR-015, PR-020), 1 over its value-engineering target (PR-010, by $505), 10 met on paper or by design (including PR-016).

13. Corrections to earlier figures

The TRL 2 documents quoted first-order values that this calculation replaces: products 30 kg (now 32.3 kg), flake 20 kg (19.3 kg), process losses 10 kg (8.4 kg), energy about 45 kWh and 0.9 kWh/kg (42.9 kWh and 0.83 kWh/kg), peak load about 13 kW (8.35 kW maximum demand with the interlock; 13.35 kW connected), equipment $22,850 (now $24,050), operating margin about $18 and result about $0 (now $20.97 and +$1.73 in v0.1, updated below). The TRL 2 note that about 2 % of the melt feed is lost as purge did not match its own 2 kg figure; the calculation uses 3 %. R8 was reported as met at the limit because only the sorting residue was counted; counting the process losses that also go to licensed disposal, it is not met.

In v0.2 (RFE-DDR-002, 2026-09-25) the 1.5 kW fume fan changes the maximum demand from 8.35 kW (36.3 A) to 8.75 kW (38.0 A), the connected load from 13.35 kW to 13.75 kW, the equipment total from $24,050 to $24,200 and the result per shift from +$1.73 to +$1.61. Energy per shift is unchanged, because the fan input depends on the air flow and pressure, not on the motor rating.

In v0.3 (RFE-DDR-003, 2026-09-30, design for construction) the equipment total moves from $24,200 to $25,300, the result per shift from +$1.61 to +$0.73, the break-even product price from $2.45/kg to $2.48/kg and the break-even input from 98 to 99 kg per shift; the clear aisle widens from 1.52 m to 1.57 m because the bins and sorting table now stand behind the painted aisle line. The hood flows are unchanged: the press booth now has two front openings, but one sliding sash always covers one of them, and the larger opening (1.2 x 0.6 m) is the one in the calculation. The duct route is shorter than the 8 m equivalent length assumed, so the pressure figure stays conservative.

In v0.5 (approved follow-ups, 2026-10-02) the equipment total moves from $25,300 to $25,505 and the result per shift from +$0.73 to +$0.57; the break-even product price stays $2.48/kg and the break-even input 99 kg per shift. The aisle headroom is now quoted to the underside of the tray padding, 2.02 m. No requirement changes status.

In v0.6 (kit 1.9.1 upgrade and STEM check, 2026-10-25) the energy per shift moves from 42.9 to 45.6 kWh and from 0.83 to 0.88 kWh/kg, because motor and drive losses are now counted (RFE-DEC-001 D2); the energy cost from $6.43 to $6.84 per shift; the operating margin from $20.97 to $20.57; the result per shift from +$0.57 to +$0.17; the break-even product price from $2.48 to $2.49/kg; and the break-even input from 99 to 100 kg per shift.

The supply figure of 8.75 kW and 38.0 A is replaced by the current check of section 5 (block A about 55 A running), and PR-017 is added as not met. The feed slot drops from 160 to 120 mm so the ISO 13857 Table 4 distance applies (RFE-DEC-001 D3); the reach path is unchanged at 1,060 mm.